

The Galois Group as Symmetry Group of the Pandrosion Quotients

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Abstract

The Galois group of an irreducible polynomial $P \in \mathbb{Q}[z]$ acts on the Pandrosion quotients by $\sigma(Q(\alpha_i, z)) = Q(\sigma(\alpha_i), z)$: it is the **symmetry group of the quotient set** $\{Q(\alpha_k, z)\}_{k=1}^d$. The identity $P' = \sum Q(\alpha_k, z)$ (Paper 14) is the *Galois trace*, and the discriminant $D^2 = \prod P'(\alpha_k)$ (Paper 22) is the *Galois norm*. The alternating group test $\text{Gal} \subseteq A_n \iff \sqrt{D^2} \in \mathbb{Q}$ follows directly. We show that the orbit decomposition of the quotient set under subgroups of Gal corresponds to intermediate field extensions, and that the twisted traces $R_\chi = \sum \chi(\sigma_k) Q(\alpha_k, z)$ recover the Lagrange resolvents. All results are verified computationally for polynomials with Galois groups $S_3, A_3, D_4, \mathbb{Z}/2, \mathbb{Z}/4$, and $(\mathbb{Z}/2)^2$.

1 The fundamental principle

Theorem 1.1 (Galois acts on quotients). *Let $P \in \mathbb{Q}[z]$ be monic of degree d with splitting field K and roots $\alpha_1, \dots, \alpha_d$. For any $\sigma \in \text{Gal}(K/\mathbb{Q})$:*

$$\boxed{\sigma(Q(\alpha_i, z)) = Q(\sigma(\alpha_i), z).} \tag{1}$$

The Galois group permutes the set $\{Q(\alpha_1, z), \dots, Q(\alpha_d, z)\}$.

Proof. $Q(\alpha_i, z) = \prod_{j \neq i} (z - \alpha_j)$. The automorphism σ permutes $\{\alpha_j : j \neq i\}$ to $\{\alpha_{\sigma(j)} : j \neq i\} = \{\alpha_j : j \neq \sigma^{-1}(i)\}$, thus $\sigma(Q(\alpha_i, z)) = Q(\alpha_{\sigma(i)}, z)$. $\square$

2 Pandrosion identities as Galois invariants

Corollary 2.1 (Trace and norm). *The identity $P' = \sum Q(\alpha_k, z)$ is the **Galois trace** of the quotient representation:*

$$P'(z) = \text{Tr}_{\text{Gal}}(Q) := \sum_{\sigma \in \text{Gal}} \sigma(Q(\alpha_1, z)) \in \mathbb{Q}[z]. \tag{2}$$

*Likewise, the discriminant identity $D(P)^2 = (-1)^{d(d-1)/2} \prod P'(\alpha_k)$ is the **Galois norm**:*

$$D^2 = \text{Nm}_{\text{Gal}}(P'(\alpha_1)) \in \mathbb{Q}. \tag{3}$$

Corollary 2.2 (Energy is Galois-invariant). *$E_P(z) = \sum Q(\alpha_k, z)^2 \in \mathbb{Q}[z]$, as it is a symmetric function of the quotients.*

3 The alternating group test

Proposition 3.1. $\text{Gal}(P/\mathbb{Q}) \subseteq A_d$ if and only if $\sqrt{D^2} \in \mathbb{Q}$.

Numerical verification:

$P(x)$	D^2	$\sqrt{D^2} \in \mathbb{Q}?$	Gal
$x^3 - 2$	-108	No	S_3
$x^3 - 3x + 1$	81	Yes (= 9)	$A_3 = \mathbb{Z}/3$
$x^4 - 2$	-2048	No	D_4
$x^2 - 2$	8	No	$\mathbb{Z}/2$
$x^4 + 1$	256	Yes (= 16)	$(\mathbb{Z}/2)^2$

4 Field extensions and quotient coefficients

Proposition 4.1. Let $P \in \mathbb{Q}[z]$ be irreducible and α a root. Then $Q(\alpha, z) \in \mathbb{Q}(\alpha)[z]$ and the coefficients of $Q(\alpha, z)$ generate $\mathbb{Q}(\alpha)$.

Proof. $Q(\alpha, z) = P(z)/(z - \alpha)$. By polynomial division in $\mathbb{Q}(\alpha)[z]$, the quotient has coefficients that are polynomials in α with rational coefficients. Since P is irreducible, these polynomials generate $\mathbb{Q}(\alpha)$. $\square$

Examples.

- $P = x^3 - 2$: $Q(\sqrt[3]{2}, z) = z^2 + \sqrt[3]{2}z + \sqrt[3]{4}$. Coefficients $\{1, \sqrt[3]{2}, \sqrt[3]{4}\}$ generate $\mathbb{Q}(\sqrt[3]{2})$ of degree 3.
- $P = x^4 - 2$: $Q(\sqrt[4]{2}, z) = z^3 + \sqrt[4]{2}z^2 + \sqrt{2}z + (\sqrt[4]{2})^3$. Coefficients generate $\mathbb{Q}(\sqrt[4]{2})$ of degree 4.

5 Orbit decomposition and intermediate fields

Theorem 5.1 (Orbit correspondence). Let $H \leq \text{Gal}(K/\mathbb{Q})$ be a subgroup fixing an intermediate field L . Then:

$$\sum_{\alpha \in H\text{-orbit}} Q(\alpha, z) \in L[z]. \quad (4)$$

The orbit sum is L -rational.

Proof. The sum $\sum_{\sigma \in H} Q(\sigma(\alpha), z)$ is fixed by every element of H , hence lies in $L[z] = K^H[z]$. $\square$

This corresponds to the factorisation of P over L : if $P = f_1 \cdots f_r$ in $L[z]$, each factor f_i contributes a Galois orbit of quotients, and $f'_i = \sum_{\alpha \in \text{roots}(f_i)} Q(\alpha, z)$ is the per-factor Pandrosion trace.

6 Twisted traces and Lagrange resolvents

Definition 6.1 (Pandrosion resolvent). For a character $\chi : \text{Gal} \rightarrow \mathbb{C}^*$:

$$R_\chi(z) := \sum_{k=1}^d \chi(\sigma_k) Q(\alpha_k, z). \quad (5)$$

Remark 6.2. $R_1(z) = P'(z)$ (trivial character = trace). For cyclic groups, $\chi_m(\sigma^k) = \omega^{mk}$ (ω a root of unity), and R_m is a polynomial over $\mathbb{Q}(\omega)$. The norm $\text{Nm}(R_m) \in \mathbb{Q}[z]$ is the classical resolvent polynomial.

7 Conclusion

Galois theory	Pandrosion
Galois group	Permutation group of $\{Q(\alpha_k, z)\}$
Galois trace	$P' = \sum Q$ (Paper 14)
Galois norm	$D^2 = \prod P'(\alpha_k)$ (Paper 22)
A_n test	$\sqrt{D^2} \in \mathbb{Q}$?
Intermediate field L	Orbit sum $\sum_{\text{orbit}} Q \in L[z]$
Character twist χ	Pandrosion resolvent $R_\chi = \sum \chi Q$
Field $\mathbb{Q}(\alpha)$	Generated by coefficients of $Q(\alpha, z)$

The Pandrosion quotient $Q(\alpha, z)$ is the natural ‘one-root window’ into the splitting field: its coefficients generate $\mathbb{Q}(\alpha)$, its Galois orbit generates all of $\{Q_k\}$, and the classical invariants (trace, norm, discriminant, resolvents) are elementary operations on the quotient set.

References

- [1] S. Lang, *Algebra*, 3rd ed., Springer (2002).
- [2] D. A. Cox, *Galois Theory*, 2nd ed., Wiley (2012).
- [3] I. Besevic, *Gauss–Lucas via Pandrosion*, Preprint (2026).
- [4] I. Besevic, *The Pandrosion discriminant*, Preprint (2026).